Moments & fluid pressure

Turn about pivots

1 Moment

Opposite moments about pivot

M = Fd, in Nm; d is perpendicular pivot-to-force-line distance.

20 N × 0.30 m = 6 Nm. Longer lever → less force for the same turning effect.

2 Balance, levers & gears

Clockwise moments = anticlockwise moments. 30 N at 0.20 m balances 15 N at 0.40 m.

Adjacent gears rotate oppositely. Larger driven gear turns more slowly with greater turning effect in an ideal system.

Apply pressure

3 Pressure on an area

Fluid = liquid or gas. Forces are normal to surfaces. **p = F/A,** Pa = N/m^2.

Smaller area → greater p for same F. **1 cm^2 = 10^{-4} m^2.**

4 Pressure with depth

p = hρg for a liquid column; increases with depth and density.

2 m water at g = 9.8 → 19,600 Pa above surface pressure. Δp = Δhρg; add atmospheric p if total requested.

Explain floating

5 Upthrust

Greater pressure on bottom than top gives upward resultant.

Floating at rest: **upthrust = weight.** Compare average density including air spaces; hollow steel ships can float.

6 Atmosphere

Air molecules collide with surfaces. Higher altitude has less air above, lower density and lower pressure.

For submerged objects, compare weight/upthrust and include other forces if present; sinking does not mean zero upthrust.